

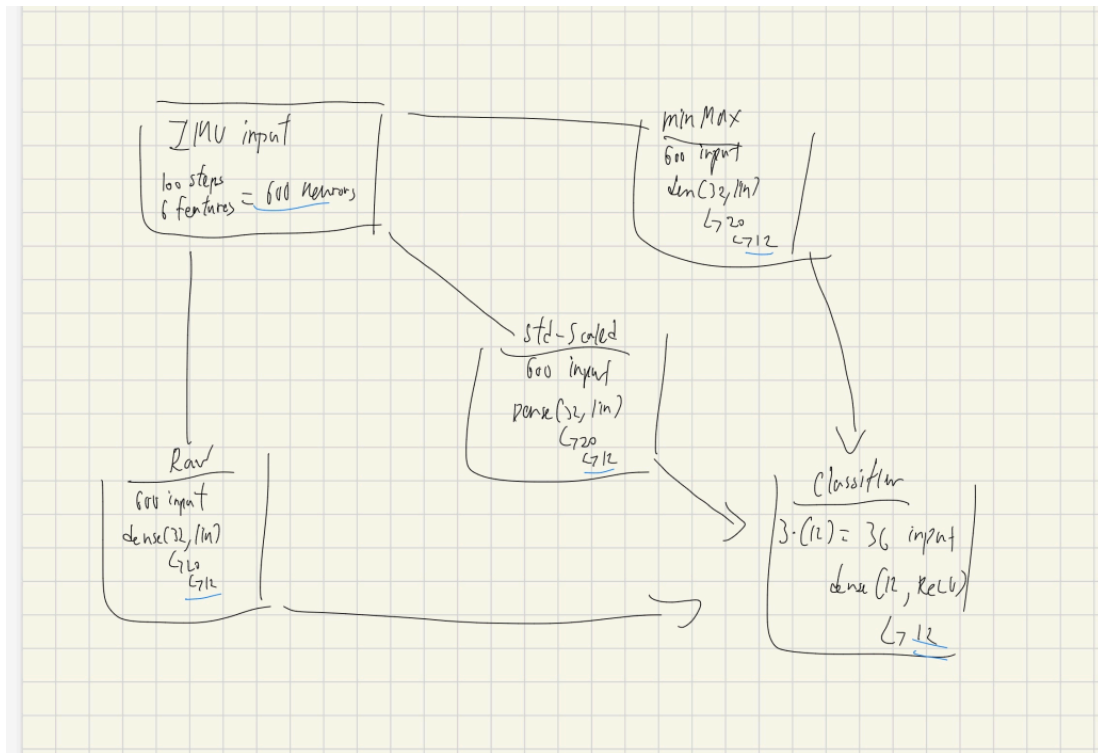
PART 1

<https://studio.edgeimpulse.com/public/1015631/live>

```
Sample length: 2000 ms.
No. of classes: 2
Starting inferencing, press 'b' to break
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 34 ms, inference 535 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 34 ms, inference 537 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 34 ms, inference 539 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  idle: 0.996094
  lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 30 ms, inference 530 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  idle: 0.039063
  lift: 0.960937
Starting inferencing in 2 seconds...
```

PART 2

Q1



Q2

Pruning is applied by removing the insignificant weights by making their weight to be 0. The sparsity goes from 10% to 80% over 500 iterations to reduce the model size. The wrappers are stripped so that it could be used as a normal Keras Model with the sparse weights. QAT helps the model adapt to int8 precision before deployment. Sample data is used during TFLite conversion to determine quantization parameters. In conclusion, Pruning and quantization reduce memory usage and improve performance on TinyML devices.

Q3

```

samples collected: 75
samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.9648, 0.0000, 0.0000, 0.0
Min-max-scaled model scores: 0.1289, 0.0273, 0.1953, 0.1953, 0.2109, 0.0000, 0.1172, 0.00
Meta-classifier scores: 0.0000, 0.1250, 0.0000, 0.0000, 0.8672, 0.0000, 0.0000, 0.0000, 0
-----
Final prediction: Climbing stairs
Class index: 4
Confidence score: 0.8672
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samples collected: 25
samples collected: 50
samples collected: 75
samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.5000, 0.0000, 0.0000, 0.0
Min-max-scaled model scores: 0.0469, 0.0000, 0.1523, 0.1523, 0.1680, 0.0117, 0.0039, 0.35
Meta-classifier scores: 0.0000, 0.5078, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0
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Final prediction: Sitting and relaxing
Class index: 1
Confidence score: 0.5078

```

When lying flat, the Arduino mostly predicted a stationary or low-motion activity class. However small fluctuations can result from sensor noise, vibrations, or quantization error for predictions.

For improvement, I would collect more data to understand the thresholds that are needed for each (or combined) metrics that would have smoothing such as majority voting across multiple outputs.

Q4

The model may repeatedly predict one class if deployment data differs from training data. Holding the board differently despite having the same intent changes IMU readings and can affect predictions. Mismatched preprocessing or feature scaling can reduce accuracy. Additionally, Int8 quantization introduces small precision losses compared to float models. Insufficient training diversity can cause poor generalization during deployment.

In conclusion, I would collect more representative data and apply consistent preprocessing and smoothing.